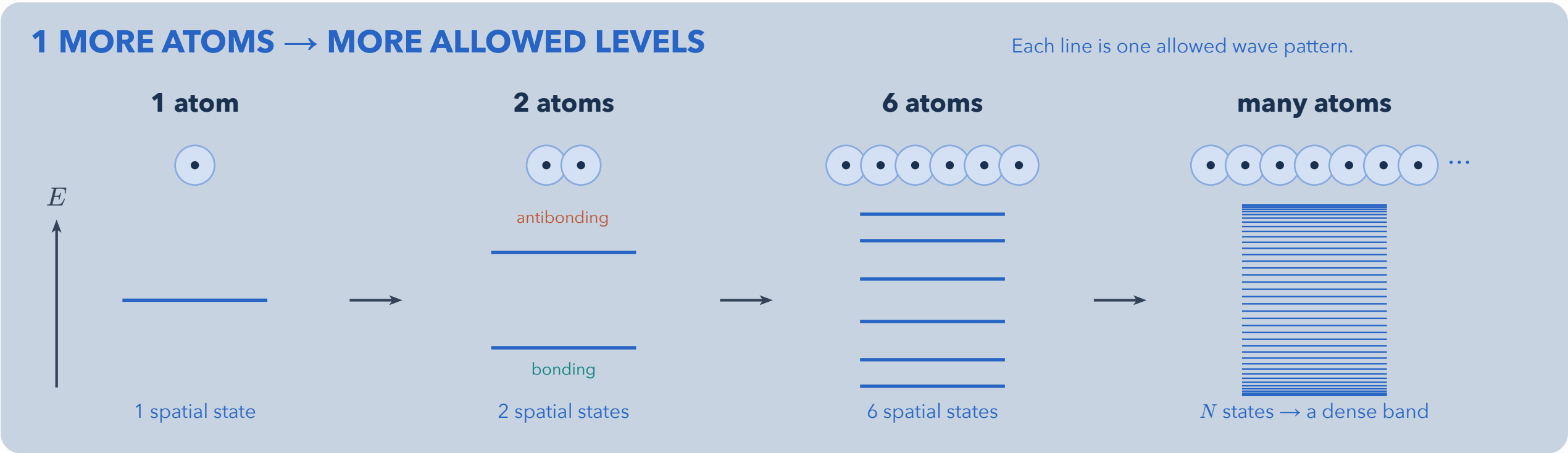


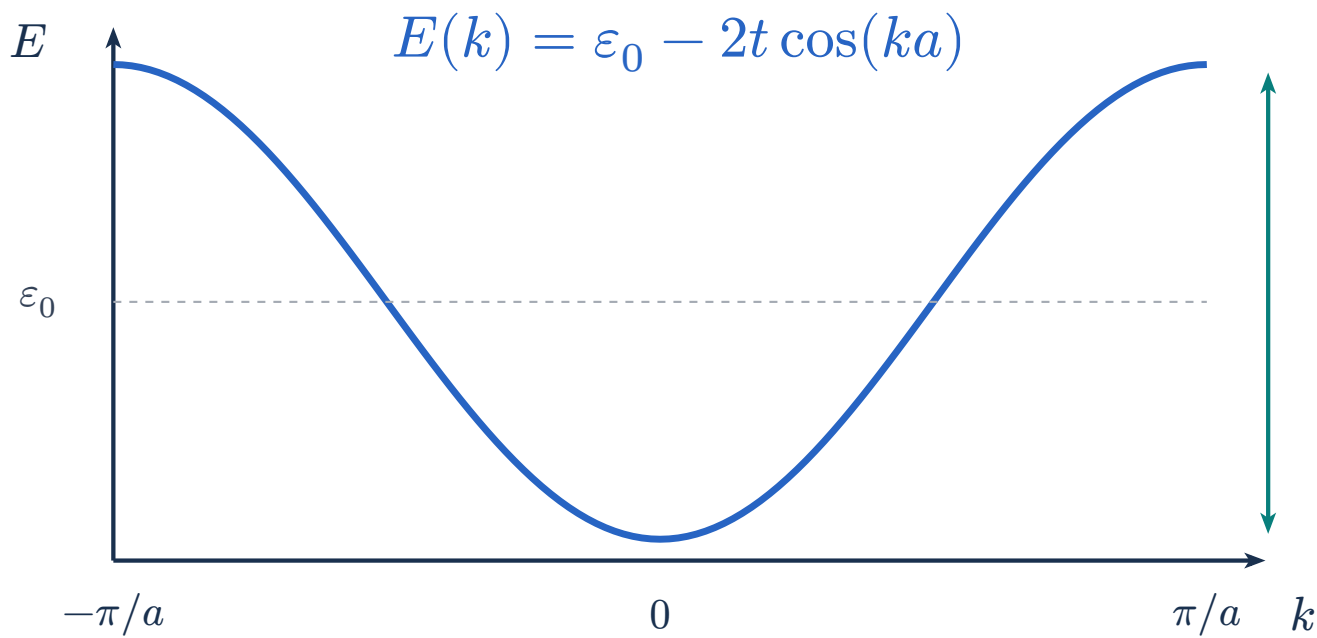
Atomic orbitals combine into electron waves spread across the solid.
Follow one orbital per atom: coupling creates the energy spread; electron filling decides what the solid can do.



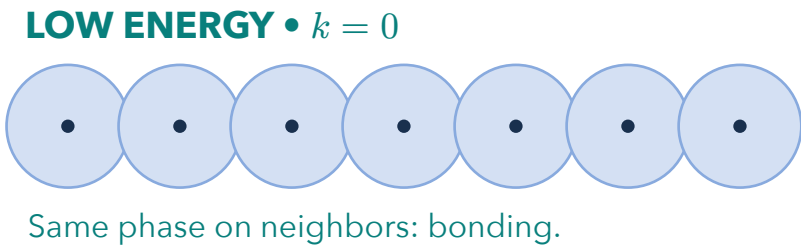
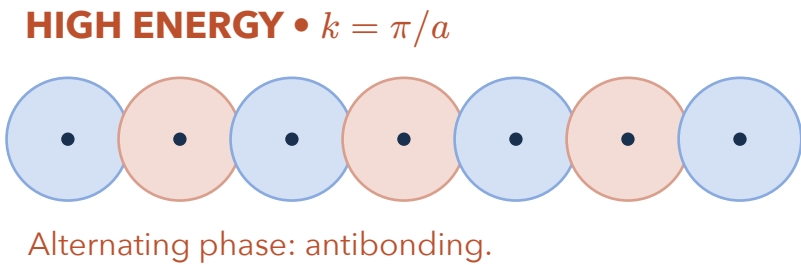
N atomic orbitals → N spatial states → room for $2N$ electrons.
Two opposite spins fit in each spatial state. Pauli exclusion controls occupation; coupling causes the splitting.

2 THE BAND IS A FAMILY OF WAVES

Minimal model: identical atoms, spacing a , one orbital each, and coupling $-t$ between neighbors ($t > 0$).



k : phase change per distance; ε_0 : isolated orbital energy.
This interval is one Brillouin zone: distinct lattice phases.



Blue / orange = wavefunction sign, not charge. Each pattern extends across the chain.

